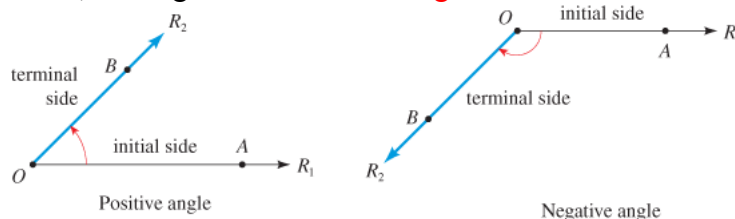


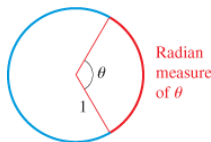
An **angle** consists of two rays with a common vertex. We often interpret an angle as a rotation of the ray R_1 onto R_2 . In this case, R_1 is called the initial side, and R_2 is called the terminal side of the angle. If the rotation is counterclockwise, the angle is considered **positive**, and if the rotation is clockwise, the angle is considered negative.



I. Angle Measure

The **measure** of an angle is the amount of rotation about the vertex required to move R_1 onto R_2 . One unit of measurement for angles is the **degree**. An angle of measure 1 degree is formed by rotating the initial side $1/360$ of a complete revolution.

Definition of Radian Measure: If a circle of radius 1 is drawn with the vertex of an angle at its center, then the measure of this angle in **radians** (abbreviated rad) is the length of the arc



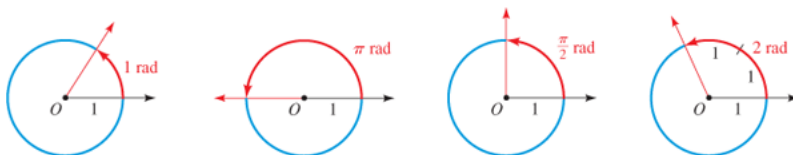
that subtends the angle.

For a circle of radius 1, a complete revolution has measure 2π rad; a straight angle has measure π rad; and a right angle has measure $\pi/2$ rad.

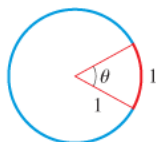
Relationship Between Degrees and Radians

To convert degrees to radians, multiply by $\pi/180$. To convert radians to degrees, multiply by $180/\pi$.

$$180^\circ = \pi \text{ rad} \quad 1 \text{ rad} = \left(\frac{180}{\pi}\right)^\circ \quad 1^\circ = \frac{\pi}{180} \text{ rad}$$



$$1 \text{ rad} \approx 57.296^\circ \quad \text{and} \quad 1^\circ \approx 0.01745 \text{ rad}$$

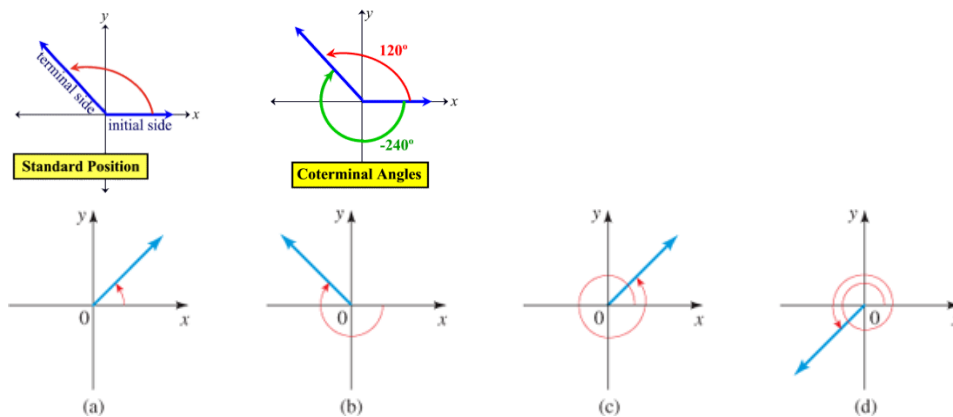


Measure of $\theta = 1 \text{ rad}$
Measure of $\theta \approx 57.296^\circ$

When no unit is given, the angle is assumed to be measured in radians.

Example 1: Expressa) 1000° in radiansb) -3.5 rad in degreesc) $\frac{5\pi}{6}$ rad in degrees.**II. Angles in Standard Position**

An angle is in **standard position** if it is drawn in the xy -plane with its vertex at the origin and its initial side on the positive x -axis. Two angles in standard position are **coterminal** if their sides coincide.

**Example 2: Coterminal Angles**

A. The measure of an angle in standard position is given. Find two positive angles and two negative angles that are coterminal with the given angle.

i. 135° ii. $\frac{11\pi}{6}$

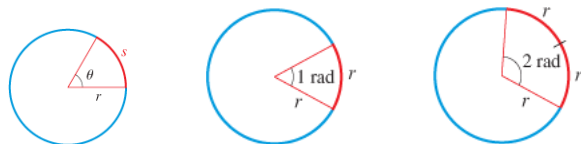
B. The measures of two angles in standard position are given. Determine whether the angles are coterminal.

i. 50° , 340° ii. $\frac{32\pi}{3}$, $\frac{11\pi}{3}$ **III. Length of a Circular Arc**

An angle whose radian measure is θ is subtended by an arc that is the fraction $\frac{\theta}{2\pi}$ of the circumference of a circle. Hence, in a circle of radius r , the length s of an arc that subtends a central angle of θ radians

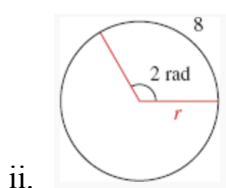
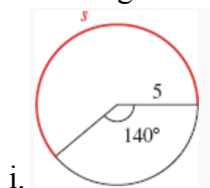
$$s = \frac{\theta}{2\pi} \times \text{circumference of circle} = \frac{\theta}{2\pi}(2\pi r) = r\theta, \underline{s = r\theta} \text{ or, solving for } \theta, \text{ we get } \underline{\theta =}$$

s/r . This formula allows us to define radian measure using a circle of radius r : The radian measure of an angle θ is the number of radiuses that can fit in the arc that subtends θ .



Example 3:

- a. Find the length s of the circular arc, the radius r of the circle, or the central angle, as indicated.



\

- b. Find the length s of the arc that subtends a central angle of 40° measure in a circle of radius 12 m.

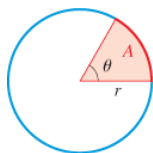
- c. A central angle θ in a circle of radius 9 m is subtended by an arc of length 14 m. Find the measure of θ in degrees and radians.

- d. Find the radius r of the circle if an arc of length 15 m on the circle subtends a central angle of $\frac{5\pi}{6}$.

IV. Area of a Circular Sector

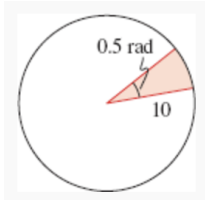
The area of a circle of radius r is $A = \pi r^2$. A sector of this circle with central angle θ has an area that is the fraction $\frac{\theta}{2\pi}$ of the area of the entire circle. Therefore, in a circle of radius r , the area A of a sector with a central angle of θ radians is

$$A = \frac{\theta}{2\pi}(\text{area of circle}) = \frac{\theta}{2\pi}(\pi r^2) = \frac{1}{2}r^2\theta. \quad \mathbf{A = \frac{1}{2}r^2\theta}$$

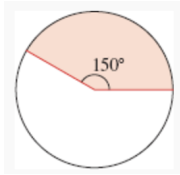


Example 4:

- a. Find the area of the sector shown in each figure.



- b. Find the radius of each circle if the area of the sector is 12.



- c. The area of a circle is 600 m^2 . Find the area of a sector of this circle that subtends a central angle of 3 rad.
- d. A sector of a circle of radius 80 mi has an area of 1600 mi^2 . Find the central angle (in radians) of the sector.

V. Circular Motion

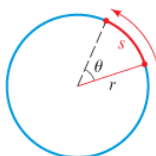
There are two ways to describe the motion of a point on a circle.

Linear speed is the rate at which the distance traveled is changing, so linear speed is the distance traveled divided by the time elapsed. **Angular speed** is the rate at which the central angle θ is changing, so angular speed is the number of radians this angle changes divided by the time elapsed.

Suppose a point moves along a circle of radius r and the ray from the center of the circle to the point traverses θ radians in time t . Let $s = r\theta$ be the distance the point travels in time t . Then the speed of the object is given by

Angular speed: $\omega = \frac{\theta}{t}$

Linear speed: $v = \frac{s}{t}$



(The symbol ω is the Greek letter “omega.”)

Relationship Between Linear and Angular Speed

If a point moves along a circle of radius r with angular speed ω , then its linear speed v is given by

$$v = r\omega$$

Example 5: A car’s wheels are 28 in. in diameter. How far (in miles) will the car travel if its wheels revolve 10,000 times without slipping?

Example 6: How many revolutions will a car wheel of diameter 30 in. make as the car travels a distance of one mile?

Example 7: A ceiling fan with 16-inch blades rotates at 45 rpm.
(a) Find the angular speed of the fan in rad/min.

(b) Find the linear speed of the tips of the blades in in./m.